Capstone Project
Weekly Meal Planner & Grocery Generator

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SECTION – 2
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Weekly Meal Planner & Grocery Generator

A simple Java application for planning meals and organizing grocery shopping.

PROJECT IDEA

The application helps users plan meals for each day of the week

Based on the selected meals , the system automatically prepares one combined grocery list.



Meal Planning

Ingredient Tracking

Auto Grocery List

MEAL PLANNING AND GROCERY LIST



Weekly Meal Planner

Meal	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Breakfast							
Lunch							
Dinner							
Snacks							



Grocery List

Category	Item	Quantity	Notes	Got it?
Produce	Apples	5	Granny Smith	
	Spinach	1 bag		
	Onions	2	Yellow	
	Garlic	1 head		
Dairy & Eggs	Milk	1 gallon	2%	
	Eggs	1 dozen	Large	
	Yogurt	6 pack	Greek, plain	
Meat & Fish	Chicken Breast	2 lbs	Boneless, skinless	
	Salmon Fillet	1 lb		
Pantry	Olive Oil	1 bottle	Extra virgin	
	Pasta	1 box	Penne	
	Canned Tomatoes	2 cans	Diced	
	Rice	1 bag	Basmati	

PROJECT ABSTRACT

- The Weekly Meal Planner and Grocery Generator is a Java-based application designed to simplify weekly meal planning and grocery management. Users can select or enter meals for each day of the week, including breakfast, lunch, dinner and snacks.

- Each meal is linked with its required ingredients and quantities. After the weekly plan is completed, the application collects the ingredients, combines duplicate items and prepares a consolidated grocery list. The list can be organized for convenient shopping and updated whenever the meal plan changes.

- The project demonstrates Java concepts such as classes and objects, methods, collections, loops, conditional statements and exception handling. The main goal is to save planning time, reduce forgotten ingredients and make grocery shopping more organized.

Introduction

This Java console application helps organize a seven-day meal plan and calculate the ingredients needed for the selected breakfast, lunch and dinner.

Project Purpose

The program combines repeated ingredients into one grocery list, so users can see the total quantity needed instead of tracking each meal separately.

Java Context

The code uses Scanner input, selection and loops, overloaded methods, 1D and 2D arrays, a Recipe class, ArrayList, HashMap, string operations, file streams and exception handling.

Problem It Solves

It reduces manual tracking and helps users avoid overlooking ingredients when preparing a shopping list.

Current Status

The current implementation supports viewing the weekly plan, viewing one day, changing a meal, generating and searching the grocery list, and saving or loading the plan.



Project Overview: Streamlining Your Kitchen



What It Does

Users can view all seven days or one day, change a meal from the available recipes, and generate a combined grocery list from the updated plan.



Why It Matters

Repeated ingredients are added together, making the list easier to use for shopping and reducing manual calculation.



Java Fundamentals

The project applies the Java topics used in the code: arrays, loops, conditions, switch, methods, objects, collections, string handling, file I/O and exception handling.

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Project Objectives

1 **Multi-Category Meal Planning**

Plan breakfast, lunch and dinner for each of the seven days of the week.

2 **Automated Grocery Lists**

Calculate ingredient quantities from the recipes in the weekly plan and combine repeated ingredients.

3 **Java Concept Demonstration**

Demonstrate relevant Java concepts from Modules 1–6 through a practical console application.

4 **User-Friendly Interface**

Provide menu options to view, search, update, save and load meal-plan information.



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Advantages



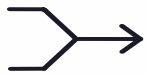
Saves Time

Automates grocery list generation from meal selections, eliminating the need for manual ingredient tracking.



Reduces Forgotten Items

Systematic ingredient tracking ensures nothing is missed from your shopping list before you head to the store.



Consolidates Duplicates

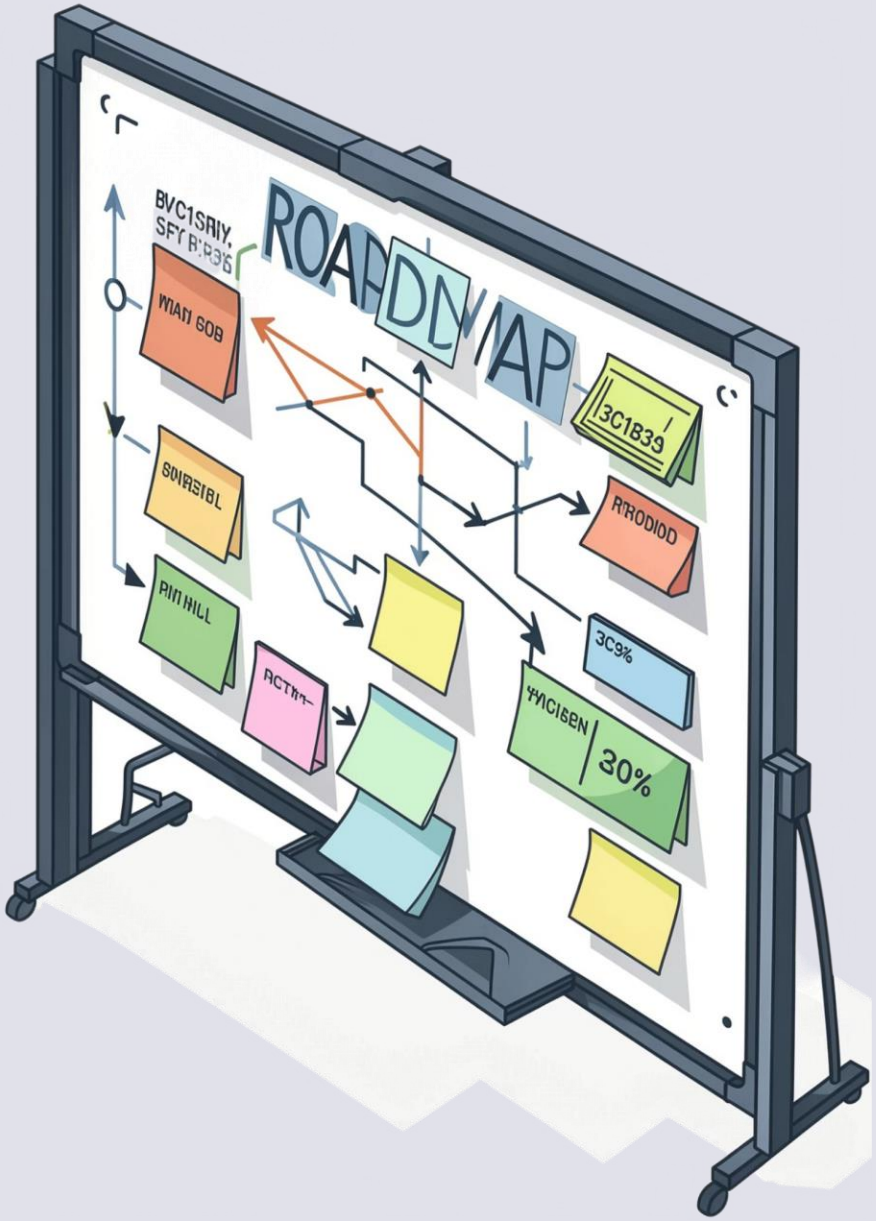
Intelligently combines repeated ingredients across multiple meals into a single, efficient shopping entry.



Flexible Planning

Supports easy updates and modifications to the weekly meal plan at any stage of the planning process.

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Future Scope

1

Database Integration

Add persistent storage for meals, ingredients, and user data.

2

Web Interface

Develop a modern web front-end for easier access and management.

3

User Accounts

Enable authentication, multi-user support, and personalized profiles.

4

Smart Nutrition Features

Include nutritional analysis, recipe suggestions, dietary preference filters, budget tracking, cost estimation, mobile app development.

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Project Development



Language & Tools

- **Language: Java**
- **IDE: Visual Studio Code**
- **Input: Scanner for menu choices and user responses**
- **File handling: FileInputStream and FileOutputStream**

Methodology & Components

- **Data storage:** 1D arrays for days and meal names; a 2D array for the seven-day meal plan
- **Program flow:** if statements, switch, for loops and a do-while menu
- **Modularity:** methods for display, searching, grocery generation and file operations
- **Object modelling:** Recipe objects hold recipe names and ingredient quantities
- **Collections:** ArrayList stores recipes; HashMap stores ingredient quantities
- **Reliability:** try-with-resources and catch blocks handle file errors

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Course Outcomes Covered



CO1 — Foundation

CO1 — Java Foundations
The program uses a Java class and main method, variables, primitive data types, literals, operators, console output and Scanner input. The meal-planning problem is broken into clear steps.



CO2 — Control Flow

CO2 — Selection & Iteration
The menu uses switch and a do-while loop. if statements validate day, meal-slot and recipe choices. for loops traverse days, meals, recipes and ingredients; break exits a switch case or search loop.



CO3 — Methods & Arrays

CO3 — Methods, Searching & Arrays
Methods divide the work into smaller tasks. displayMealPlan() is overloaded to show the whole week or one day. The program traverses 1D arrays and the 7×3 weeklyPlan array, and searches recipes and grocery items.

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Course Outcomes Covered: CO4–CO6

4

CO4 — Classes, Objects & Exceptions

Recipe is a class; each recipe is an object created with new. Its fields store a name and ingredient quantities, and a constructor initializes the name and ingredient map. try-with-resources and catch handle file errors.

5

CO5 — Strings & File I/O

String methods such as trim() and equalsIgnoreCase() support searching and validation. split() parses saved records and StringBuilder assembles file content. FileInputStream and FileOutputStream load and save the plan.

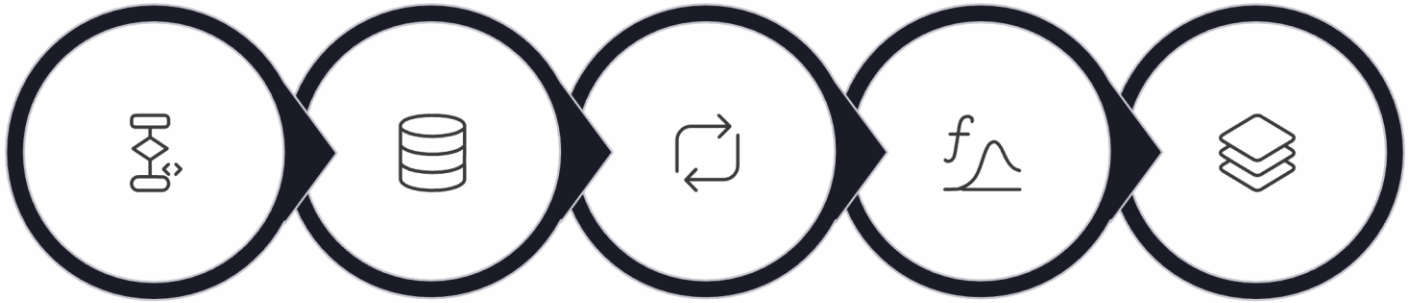
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CO6 — Collections & Generics

ArrayList<Recipe> stores available recipes. HashMap<String, Integer> stores ingredient quantities. Generic types specify the data held by each collection, and for-each loops traverse collection entries.

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Topics Covered



Algorithm Data Types Loops Methods Overloading

Java Topics Applied (CO1–CO6)

CO1 — Variables, data types, literals, operators, Scanner input and console output

CO2 — if statements, switch, for loops, do-while, nested loops, validation and break

CO3 — Methods, return values, method overloading, enhanced for loop, array traversal and searching, 1D and 2D arrays

CO4 — Recipe class, objects, fields, constructor, static members and exception handling

CO5 — String comparison and trimming, split(), StringBuilder, file reading/writing and try-with-resources

CO6 — ArrayList, HashMap, generics, lookup and for-each iteration

Topics are selected from the supplied Modules 1–6 syllabus and included only where the current project demonstrates them.

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THANK YOU

